



SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)



SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA1402

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS:

Total Marks:50

Annexure A

SCENARIO BASED TASK

Code of Conduct:

I K.Manoj Kumar Reddy(192425266) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.Manoj

1. Parsing table conflict in LL(1)

Answer:

* In an LL(1) parser, the syntax analyzer constructs the parsing table using the FIRST and FOLLOW sets.

* FIRST set determines the first terminal that can appear from a production.

* FOLLOW set determines the terminals that can appear immediately after a non-terminal.

A conflict occurs when than one production is entered in the same parsing table all this usually happens because:

* The grammar is left recursive or

* Two production have common prefixes, making their FIRST sets overlap.

Due to this conflict, the parser cannot uniquely decide which production to use hence, the grammar is not deterministic and is not LL(1).

Decision: The grammar should be modified by eliminating left recursion and performing left factoring so that each parsing table entry contains only one production, making the grammar suitable for LL(1) parsing.

2. Incorrect parse tree Generation:

Answer:

* The syntax analyzer constructs a parse tree according to the grammar rules.

* For expression $id + id - id$, an incorrect parse tree may be generated if the grammar does not enforce left associativity. The expression may be grouped as:

$\rightarrow id + (id - id);$

instead of:

$\rightarrow (id + id) - id;$

This production produces incorrect evaluation.

* The grammar determines how the parse tree is formed if associativity is not specified, different parse trees can be generated

Decision: The grammar should be rewritten to enforce left associativity or precedence and associativity rules should be affixed in the parser to ensure correct parsing and evaluation.

3. Invalid expression handling.

Answer:

* The syntax analyzer reviews the token sequence; a+b
* Although each token is valid, the sequence does not follow grammar rule

* After the operator +, the parser expects an operand, but another operator is found * therefore, no valid parse tree can be generated

* In predictive parsing, this parser reports an error because no production matches the input

* In shift-reduce parsing, no valid shift or reduce action is possible, resulting in a syntax error.

Decision: The parser should implement error recovery techniques such as panic mode recovery techniques such as panic mode recovery or phrase level recovery to report meaningful errors and continue parsing the remaining input

4. Ambiguity in conditional statement

Answer:

* Given the grammar,

$\rightarrow S \rightarrow \text{if } E \text{ then } S$

$\rightarrow S \rightarrow \text{if } E \text{ then } S \text{ else } S$

$\rightarrow S \rightarrow \epsilon$

* The syntax analyzer faces ambiguity because it cannot determine which production to apply when it encounters else

* This ambiguity problem is known as the Dangling else problem.

* The ambiguity causes conflicts during parsing because the parser cannot determine whether the else belongs to the nearest if or an earlier if

Decision: The grammar should be modified to distinguish matched and unmatched statements, or the parser should follow the rule that else is associated with the nearest unmatched if. This removes ambiguity and ensures correct parsing.

5. Operator Precedence conflict.

Answer:

* Given the grammar

$\rightarrow E \rightarrow E + E$

$\rightarrow E \rightarrow E * E$

$\rightarrow E \rightarrow \text{id}$

* Consider where parsing the input:

$\rightarrow \text{id} + \text{id} * \text{id}$

* The grammar is ambiguous because it does not specify operator precedence

* The parser may generate either

$\rightarrow id + id) * id$

or

$\rightarrow id + (id * id)$

* leading to incorrect evaluation

* Since, multiplication has higher precedence than addition, the correct parse tree is

$\rightarrow id + (id * id)$

* Associative should be defined so that operator with the same precedence are evaluated correctly.

Decision: Rewrite the grammar as:

$\rightarrow E \rightarrow E + T /$

$\rightarrow T \rightarrow T * F /$

$\rightarrow F \rightarrow id$

Or define precedence and associativity rule



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Engineer to Excel

RUBRICS FOR CALCULATION:

Scenario based assessment

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and